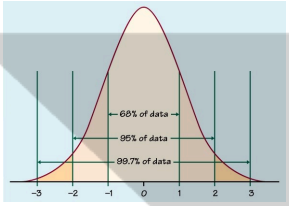
1. The Normal Distribution

1.1. Introduction

- A normal distribution is used to model a **continuous** random variable, thus it is a **continuous distribution** and not a discrete distribution. It has two parameters, the mean (μ) and the variance (σ^2).

A continuous random variable that follows a normal distribution, $N(\mu, \sigma^2)$ can be written as follow

$$X \sim N(\mu, \sigma^2)$$



- The area under the graph represents probability
- The mean (μ) acts as a line of symmetry
  - 50% of the probability lies on the right side and 50% of the probability lies on the left side.
- The standard deviation represents how spread out a value is from the mean.

1.2. Standardising a Normal Distribution

- To convert any value in the normal distribution $X \sim N(\mu, \sigma^2)$ to the standard normal distribution $N(0,1)$, we use the **standardization** formula
  - This is done as the z values are already given a probability value for the standard normal distribution.

The x value in the normal distribution is the z value in the standard normal distribution.

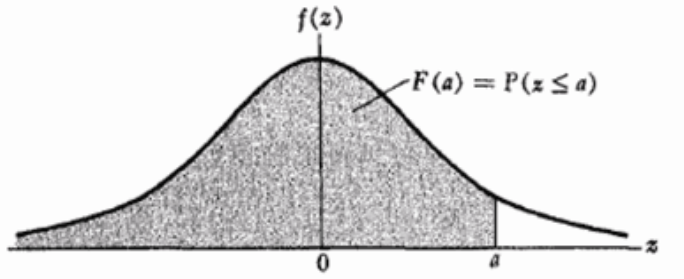
$$z = \frac{x - \mu}{\sigma}$$

- Note that you divide by the **standard deviation** and not the variance.
- It is also not required to standardise a normal distribution if the continuous random variable is already following a standard normal distribution $X \sim N(0,1)$.

1.3. Finding Probabilities

- The standard normal distribution can be used to find the probability of an event occurring.
  - The notation for the **area** to the **left** of z is $\phi(z)$.

Consider a standard normal distribution, $X \sim N(0,1)$, where you want to find $P(z \leq a)$.



We can use the standard normal table to find the value of a, and match it with the corresponding probability.

z	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
											ADD								
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	4	8	12	16	20	24	28	32	36
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753	4	8	12	16	20	24	28	32	36
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141	4	8	12	15	19	23	27	31	35
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517	4	7	11	15	19	22	26	30	34
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879	4	7	11	14	18	22	25	29	32
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224	3	7	10	14	17	20	24	27	31
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549	3	7	10	13	16	19	23	26	29
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852	3	6	9	12	15	18	21	24	27
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133	3	5	8	11	14	16	19	22	25
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389	3	5	8	10	13	15	18	20	23
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621	2	5	7	9	12	14	16	19	21
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830	2	4	6	8	10	12	14	16	18
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015	2	4	6	7	9	11	13	15	17
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177	2	3	5	6	8	10	11	13	14
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319	1	3	4	6	7	8	10	11	13
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441	1	2	4	5	6	7	8	10	11
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545	1	2	3	4	5	6	7	8	9
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633	1	2	3	4	4	5	6	7	8
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706	1	1	2	3	4	4	5	6	6
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767	1	1	2	2	3	4	4	5	5
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817	0	1	1	2	2	3	3	4	4
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857	0	1	1	2	2	2	3	3	4
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890	0	1	1	1	2	2	2	3	3
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916	0	1	1	1	1	2	2	2	2
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936	0	0	1	1	1	1	1	2	2
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952	0	0	0	1	1	1	1	1	1
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964	0	0	0	0	1	1	1	1	1
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974	0	0	0	0	0	1	1	1	1
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981	0	0	0	0	0	0	0	1	1
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986	0	0	0	0	0	0	0	0	0

- The left side of the table gives us a decimal value, and the right side of the table gives us the value we need to add to the decimal value.
  - The reverse of this process can be used to find the z value from a given area.

Let a be equal to **1.645** in this example. Using the table above, given in MF19, we can find the corresponding probability for $\phi(1.645)$ as such:

z	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
											ADD								
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	4	8	12	16	20	24	28	32	36
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1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633	1	2	3	4	4	5	6	7	8
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706	1	1	2	3	4	4	5	6	6
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767	1	1	2	2	3	4	4	5	5

Using the table:

$$\phi(1.645) = 0.9495 + 0.0005 = 0.95$$

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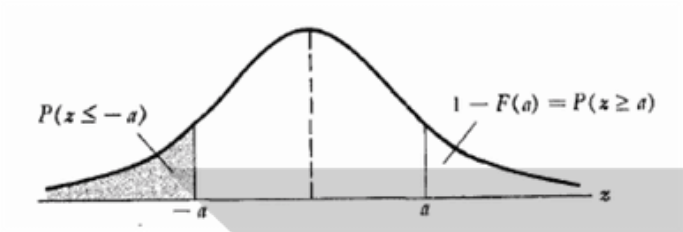
- This also represents 95% of the area under the graph.
- The **mean value** can be found by probability $\times$ sample space.

Symmetry of the Distribution

- The mean, μ, acts as a line of symmetry for the normal distribution.
- A z value less than the mean will have a **negative value**.

Given a probability $P(z \geq a)$ or $P(z \leq -a)$



We can find the value by:

$$1 - P(z \leq a)$$

Some other useful results are listed below

Probability	Area
$P(z < a)$	$\phi(a)$
$P(z > a) = 1 - P(z < a)$	$1 - \phi(a)$
$P(z > -a) = P(z < a)$	$\phi(a)$
$P(z < -a) = 1 - P(z < a)$	$1 - \phi(a)$

- It is also important to know that $P(z = a) = 0$ due to the continuous random variable being continuous.

Area in between two values

- The area in between two values can be found by subtracting the two areas.

Some useful results are listed below

Probability	Area
$P(a \leq z \leq b)$	$\phi(b) - \phi(a)$
$P(-a \leq z \leq -b)$	$\phi(a) - \phi(b)$
$P(-a \leq z \leq b)$	$\phi(b) + \phi(a) - 1$
$P(-a \leq z \leq a)$	$2(\phi(a)) - 1$

Example

For a random variable X with normal distribution

$$X \sim N(20,\ 4^2)$$

1. Find the probability of $P(X \leq 25)$
2. Find the probability of $P(X \geq 25)$
3. Find the probability of $P(X \leq 12)$
4. Find the probability of $P(10 \leq X \leq 30)$

Answer

1. Find the probability of $P(X \leq 25)$

First we standardize the probability

$$Z = \frac{25 - 20}{4} = 1.25$$

Search for this value in normal tables

$$\Phi(1.25) = 0.8944$$

2. Find the probability of $P(X \geq 25)$

Change from greater than to less than using:

$$P(Z \geq a) = 1 - P(Z \leq a)$$

$P(X \geq 25) = 1 - P(X \leq 25$)

Using the probability from above

$$P(X \geq 25) = 1 - 0.8944 = 0.1057$$

3. Find the probability of $P(X \leq 12)$

Standardize the probability

$$Z = \frac{12 - 20}{4} = -2$$

Change from negative value to positive by:

$$P(Z \leq -a) = 1 - P(Z \leq a)$$

$$P(Z \leq -2) = 1 - P(Z \leq 2)$$

Search for 2 in the normal tables

$$P(Z \leq -2) = 1 - 0.9773 = 0.0228$$

4. Find the probability of $P(10 \leq X \leq 30)$

Split inequality into two using:

$$P(a \leq Z \leq b) = P(Z \leq b) - P(Z \leq a)$$

$$P(10 \leq X \leq 30) = P(X \leq 30) - P(X \leq 10)$$

Standardize the values

$$= P(Z \leq 2.5) - P(Z \leq -2.5)$$

Convert the negative values to positive values

$$= P(Z \leq 2.5) - (1 - P(Z \leq 2.5))$$

Search for 2.5 in the normal table

$$= 0.9938 - (1 - 0.9938) = 0.9876$$

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1.4. Finding the mean and variance

Example

The lengths of body feathers of a particular species of bird are modelled by a normal distribution. A researcher measures the lengths of a random sample of 600 feathers and finds that 63 are less than 6 cm long and 155 are more than 12 cm long.

i. Find estimates of the mean and standard deviation of the lengths of body feathers of birds of this species.

ii. In a random sample of 1000 body feathers from birds of this species, how many would the researcher expect to find with lengths more than 1 standard deviation from the mean?

Answer

Part (i):

Interpreting the question and finding probabilities:

$P(X < 6) = 0.105 \quad P(X > 12) = 0.258$

For $X < 6$, the probability cannot be found on the tables which means it is behind the mean and therefore we must find $1-$ and assume probability is negative

$$-P(X < 6) = 0.895$$

Using the standardization formula and working back from the table as we are given probability

$$\frac{6-\mu}{\sigma} = -1.253$$

Convert the greater than sign to less than

$$P(X > 12) = 1 - P(X < 12)$$

$$P(X < 12) = 1 - 0.258 = 0.742$$

Work back from table and use standardization formula

$$\frac{(12-\mu)}{\sigma} = 0.650$$

Solve the simultaneous equations

$\sigma = 3.15$ and $\mu = 9.9$

Part (ii):

Greater than 1 standard deviance from μ means both sides of the graph however area symmetrical therefore find the greater area & double it

Using values calculated from (i):

$$P(X > (9.9 + 3.15) = P(X > 13.05)$$

Standardise this value

$$\frac{13.05 - 9.9}{3.15} = 1$$

We re-write the inequality as such

$$P(Z > 1) = 1 - P(Z < 1)$$

Find probability of 1 and find $P(Z > 1)$

$$P(Z > 1) = 1 - 0.841 = 0.1587$$

Multiply the probability by 2 to take into account the other side.

$$0.1587 \times 2 = 0.3174$$

Multiply the probability with the sample size

$0.3174 \times 1000 = 317$ birds

1.5. Normal approximation

- The binomial distribution can be used to approximate a normal distribution.

Let X follow a binomial distribution such that

$$X \sim B(n, p)$$

$$E(X) = \mu = np$$

$$Var(X) = \sigma^2 = np(1-p) = npq$$

Using this, we can approximate a normal distribution:

$$X \sim N(np, npq)$$

Conditions to approximate a normal distribution:

- $np > 5$
- $nq > 5$

Continuity Correction

- A continuity correction must be applied as a binomial distribution is a **discrete** distribution, while a normal distribution is a **continuous** distribution.
  - This can be thought of as making a histogram continuous by manipulating an interval's lower and upper bounds. Due to this, we must **add** or **subtract 0.5** according to the inequality in the probability.

$$z = \frac{x \pm 0.5 - \mu}{\sigma}$$

Consider a constant a

Probability	Continuity correction	Notes
$P(x > a)$	$P(x > a + 0.5)$	Does not include a
$P(x \geq a)$	$P(x > a - 0.5)$	Includes a
$P(x < a)$	$P(x < a - 0.5)$	Does not include a
$P(x \leq a)$	$P(x < a + 0.5)$	Includes a
$P(a < x < b)$	$P(a + 0.5 < x < b - 0.5)$	Does not include a or b

Example

On a certain road 20% of the vehicles are trucks, 16% are buses and remainder are cars. A random sample of 125

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vehicles is taken. Using a suitable approximation, find the probability that more than 73 are cars.

Answer

First we find the probability of the cars

$$1-(0.16+0.2)=0.64$$

So the binomial distribution is

$$X \sim B(125, 0.64)$$

First we check if a normal approximation can be used*

$125 \times 0.64 = 80$ and $125 \times (1-0.64) = 45$

Both conditions are satisfied so we can approximate a normal distribution.

$$X \sim B(125,\ 0.64) \rightarrow X \sim N(80,\ 28.8)$$

Apply the continuity correction

$$P(X>73)=P(X \geq 73.5)$$

Now we can find the probability

$$P(X \geq 73.5)=1-P(X \leq 73.5)$$

Next we standardise this value

$$Z=\frac{73.5-80}{\sqrt{28.8}}=-1.211$$

As it is a negative value, we must subtract by one again

$$1-(1-P(Z<-1.211)=P(Z<1.211)=\phi(1.211)$$

Using the normal table

$$P=0.8871$$

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